Kinematics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

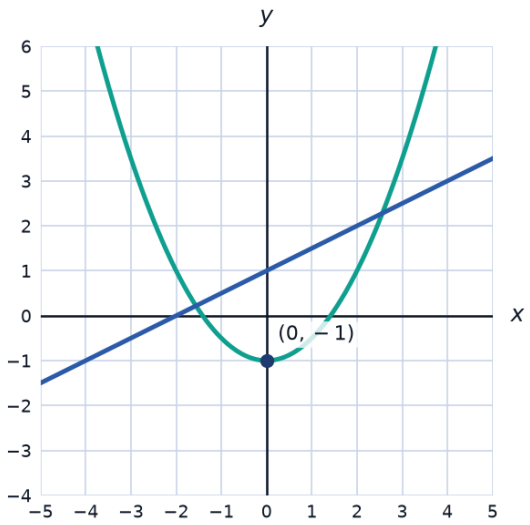
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Displacement, velocity, and acceleration**

A particle moves with displacement $s = t^3 - 6t^2 + 9t$ (in meters, t in seconds). Find expressions for its velocity v and acceleration a.

2 **[Foundational]** • **Target: Displacement, velocity, and acceleration**

A particle moves with displacement $s = 2t^3 - 9t^2 + 12t$ (in meters, t in seconds). Find its velocity at $t = 2$ seconds.



3 **[Foundational]** • **Target: The suvat equations for constant acceleration**

A car starts from rest and accelerates uniformly at 2.5 m/s^2 for 8 seconds. Find its final velocity and the distance traveled, using $v = u + at$ and $s = ut + \frac{1}{2}at^2$.

4 **[Foundational]** • **Target: The suvat equations for constant acceleration**

A ball is thrown upward with initial velocity $u = 15$ m/s. Using $g = 9.8$ m/s^2 (deceleration), find the maximum height reached, using $v^2 = u^2 + 2as$ with $v = 0$ at the top.

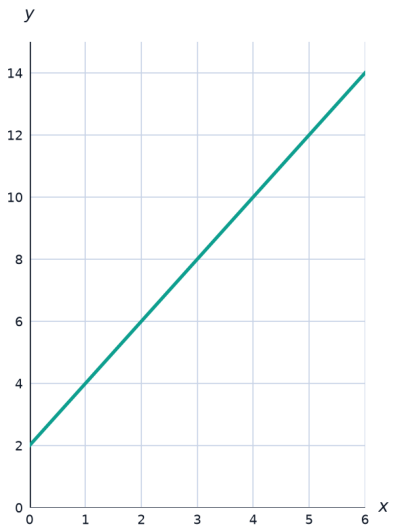
5 **[Foundational]** • **Target: The suvat equations for constant acceleration**

A cyclist decelerates uniformly from 12 m/s to rest over a distance of 36 meters. Find the deceleration, using $v^2 = u^2 + 2as$.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Velocity-time graphs

A particle accelerates uniformly from $u = 2$ m/s to $v = 14$ m/s over 6 seconds, then travels at constant velocity for a further 4 seconds. The velocity-time graph is shown. Find the total distance traveled, using the area under the graph.



7 [Foundational] • Target: Velocity-time graphs

A particle decelerates uniformly from $v = 20$ m/s to rest over 5 seconds, giving a velocity-time graph that is a straight line from $(0, 20)$ to $(5, 0)$. Find the distance traveled, using the area under the graph.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Motion under gravity

A stone is dropped from a height of 45 meters. Using $g = 9.8$ m/s^2, find the time taken to reach the ground, and its speed on impact.

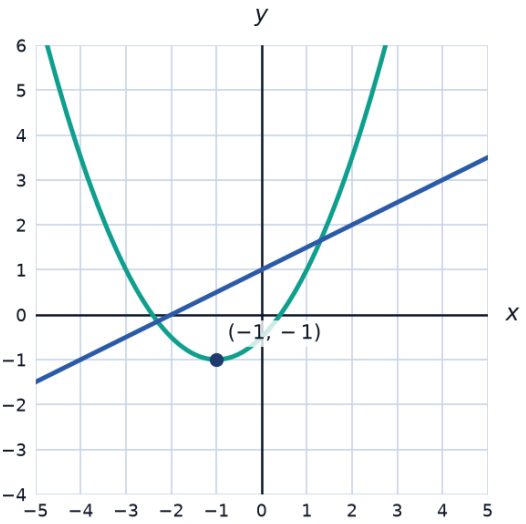
9 [Intermediate] • Target: Motion under gravity

A ball is thrown vertically upward with initial speed $u = 24.5$ m/s. Using $g = 9.8$ m/s^2, find the total time taken to return to its starting point.

© ClipSAT — clipsat.org | All Rights Reserved

10 [Intermediate] • Target: Using calculus for variable acceleration

A particle has velocity $v = 3t^2 - 2t$ (m/s). Find its displacement from $t = 0$ to $t = 3$, and its acceleration at $t = 3$.



11 [Intermediate] • Target: Using calculus for variable acceleration

A particle has velocity $v = 4t - t^2$ (m/s). Find its displacement from $t = 0$ to $t = 2$, and its acceleration at $t = 2$.

12 [Intermediate] • Target: Finding when a particle is at rest or changes direction

A particle has velocity $v = t^2 - 5t + 6$ (m/s). Find the times at which the particle is instantaneously at rest.

13 [Intermediate] • Target: Finding when a particle is at rest or changes direction

A particle has velocity $v = t^2 - 7t + 10$ (m/s). Find the times at which the particle is instantaneously at rest.

14 [Intermediate] • Target: Relative and combined motion

Two cars start 300 meters apart on a straight road and travel toward each other, one at 18 m/s and the other at 12 m/s, both at constant speed. Find the time until they meet.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

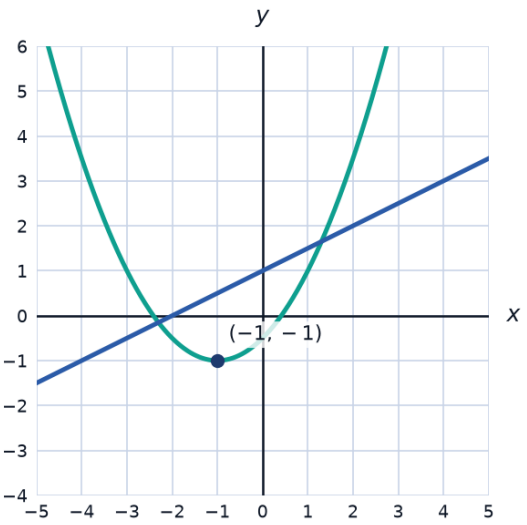
15 [Challenging] • Target: Relative and combined motion • Cross-Lesson: 8.01 Functions: Composite & Inverse

Two cars start at the same point on a straight road, one traveling at 25 m/s. The second car starts 4 seconds later, traveling at 30 m/s. Find how long after the first car starts the second car catches up.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: A well-designed word problem: a train between two stations • Cross-Lesson: 9.01 Circular Measure

This question has two parts. A train starts from rest at station A and accelerates uniformly at 0.5 m/s^2 until it reaches a speed of 20 m/s. It then travels at this constant speed before decelerating uniformly at 1 m/s^2 to come to rest at station B. The total distance between the stations is 2400 meters. (a) Find the time and distance covered during the acceleration phase. (b) Find the total journey time between the two stations.



17 [Challenging] • Target: Using calculus for variable acceleration • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude 15 N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is 120°, find the magnitude of the third force.

18 [Challenging] • Target: Finding when a particle is at rest or changes direction • Cross-Lesson: 10.01 Series & Binomial Theorem

A constant force of 15 N is applied to an object, causing it to move a distance of 4 m in the direction of the force. Calculate the work done by the force.

19 [Challenging] • Target: Relative and combined motion • Cross-Lesson: 12.01 Forces & Equilibrium

Three concurrent forces keep a particle in equilibrium, each pair separated by 120°, and two of the forces have magnitude 16 N. Find the magnitude of the third force.

20 [Challenging] • Target: A well-designed word problem: a train between two stations • Cross-Lesson: 13.01 Newton's Laws

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200. | Fertilizer Type | Yield < 50 kg | Yield $\geq$ 50 kg | Total | |---|---|---|---| | A | 30 | 70 | 100 | | B | 50 | 50 | 100 | | Total | 80 | 120 | 200 | Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.

© ClipSAT — clipsat.org | All Rights Reserved

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Constant Acceleration & SUVAT Equations

A car accelerates uniformly from rest to a speed of 24 m/s over a distance of 144 m. Find its acceleration.

a. 2 m/s^2

b. 1.5 m/s^2

c. 3 m/s^2

d. 4 m/s^2

22 [Foundational] • Target: Constant Acceleration & SUVAT Equations

A ball is thrown vertically upwards with an initial speed of 20 m/s from ground level. Taking $g = 10$ m/s^2, what is the maximum height reached?

a. 20 m

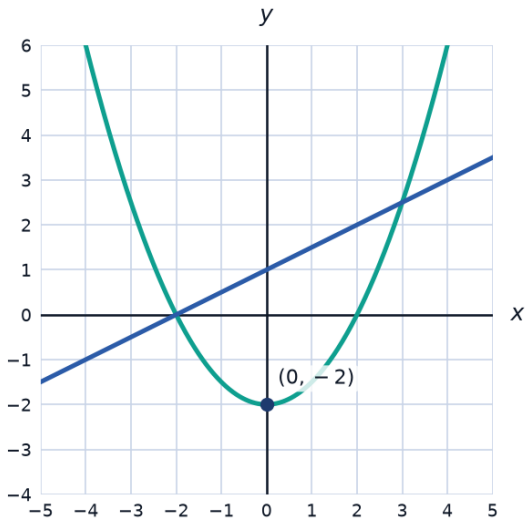
b. 40 m

c. 15 m

d. 25 m

23 [Foundational] • Target: Constant Acceleration & SUVAT Equations

A train decelerates uniformly from 30 m/s to 10 m/s in 8 seconds. How far does the train travel during these 8 seconds?



a. 160 m

b. 140 m

c. 180 m

d. 120 m

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Velocity-Time & Displacement-Time Graphs

On a velocity-time graph, what physical quantities are represented by the gradient and the area under the graph, respectively?

a. Acceleration and Displacement

b. Displacement and Acceleration

c. Speed and Distance

d. Force and Momentum

25 [Foundational] • Target: Velocity-Time & Displacement-Time Graphs

A particle moves according to the following v-t graph: accelerates from rest to 15 m/s in 6 s, travels at constant 15 m/s for 10 s, then decelerates to rest in 4 s. What is the total distance traveled?

a. 225 m

b. 250 m

c. 200 m

d. 270 m

26 [Foundational] • Target: Velocity-Time & Displacement-Time Graphs

A particle's displacement is given by $s(t) = 4t - t^2$ for $0 \le t \le 4$. At what time does the particle momentarily come to rest?

a. $t = 2$ s

b. $t = 4$ s

c. $t = 1$ s

d. $t = 0$ s

27 [Foundational] • Target: Variable Acceleration via Calculus

The velocity of a particle moving along a straight line is given by $v(t) = 6t^2 - 18t + 12$ for $t \ge 0$. Find the acceleration at $t = 3$.

a. 18 m/s^2

b. 12 m/s^2

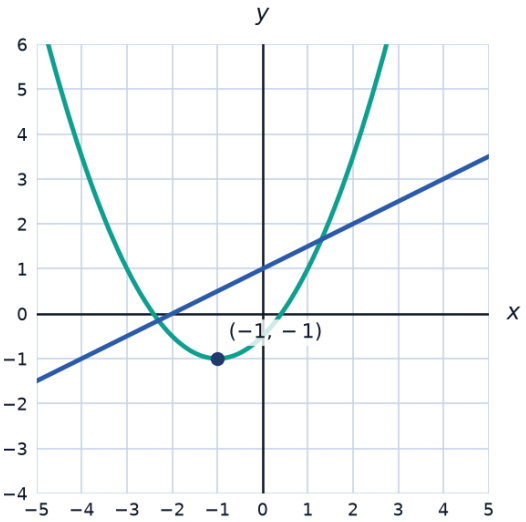
c. 24 m/s^2

d. 6 m/s^2

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Variable Acceleration via Calculus

A particle starts from the origin at $t = 0$ with velocity $v(t) = 3t^2 - 4t$. Find its displacement $s(t)$ at $t = 3$.



a. 9 m

b. 18 m

c. 15 m

d. 6 m

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Variable Acceleration via Calculus

A particle has acceleration $a(t) = 4 - 2t$ m/s^2. If $v(0) = 5$ m/s, find the maximum velocity of the particle.

a. 9 m/s

b. 7 m/s

c. 8 m/s

d. 11 m/s

30 [Intermediate] • Target: Multi-Stage Motion & Exam Challenges

A stone is dropped from the top of a cliff of height H. One second later, a second stone is thrown vertically downwards with speed 20 m/s. Both stones hit the sea at the same instant. Taking $g = 10$ m/s^2, find H.

a. 45 m

b. 80 m

c. 60 m

d. 35 m

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Multi-Stage Motion & Exam Challenges

A police car initially at rest begins accelerating at $3\ \text{m/s}^2$ the moment a speeding car passes it at a constant speed of $24\ \text{m/s}$. How long does it take the police car to overtake the speeding car?

a. 16 s

b. 12 s

c. 8 s

d. 20 s

32 [Intermediate] • Target: Multi-Stage Motion & Exam Challenges

A particle moving in a straight line passes through a fixed point O with velocity $8\ \text{m/s}$. Its acceleration is $a = -0.5v\ \text{m/s}^2$. What distance does it travel before its speed halves to $4\ \text{m/s}$?

a. 8 m

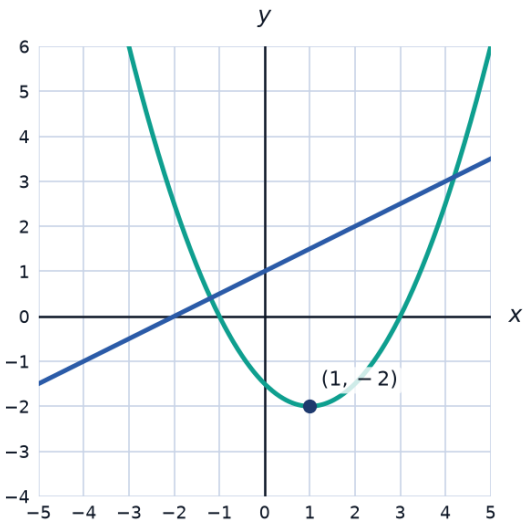
b. 4 m

c. 12 m

d. 16 m

33 [Intermediate] • Target: Displacement, velocity, and acceleration

A particle of mass 8 kg is moving under the action of a net force $\mathbf{F}$ N. The acceleration of the particle, $\mathbf{a}$ m/s^2, is given by the vector $\mathbf{a} = (2t - 4;\ t^2 - 9)$ for $t \geq 0$, where t is the time in seconds. The particle is initially at rest at the origin.



a. 8 N

b. 16 N

c. 24 N

d. 32 N

© ClipSAT — clipsat.org | All Rights Reserved

34 **[Intermediate]** • **Target: The suvat equations for constant acceleration**

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 3t - 2$. Find the displacement of the particle during the first 4 seconds of its motion.

a. 16 m

b. 20 m

c. 24 m

d. 28 m

35 **[Intermediate]** • **Target: Velocity-time graphs**

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 3t^2 - 21t + 36$ m/s. Find when the particle is momentarily at rest.

a. $t = 4$ s or $t = 4$ s

b. $t = 3$ s or $t = 5$ s

c. $t = 3$ s only

d. $t = 3$ s or $t = 4$ s

36 **[Intermediate]** • **Target: Motion under gravity**

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 3t^2 - 18t + 24$ m/s. Find when the particle is momentarily at rest.

a. $t = 3$ s or $t = 4$ s

b. $t = 2$ s only

c. $t = 2$ s or $t = 4$ s

d. $t = 2$ s or $t = 5$ s

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Using calculus for variable acceleration

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 18t + 40$ m/s. Find when the particle is momentarily at rest.

a. $t = 4$ s or $t = 5$ s

b. $t = 4$ s or $t = 6$ s

c. $t = 5$ s or $t = 5$ s

d. $t = 4$ s only

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Finding when a particle is at rest or changes direction • Cross-Lesson: 10.01 Series & Binomial Theorem

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 3t^2 - 33t + 90$ m/s. Find when the particle is momentarily at rest.

a. $t = 5$ s or $t = 6$ s

b. $t = 5$ s or $t = 7$ s

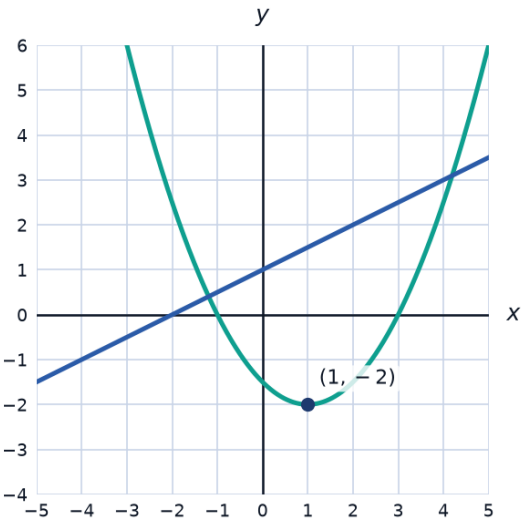
c. $t = 6$ s or $t = 6$ s

d. $t = 5$ s only

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Relative and combined motion • Cross-Lesson: 12.01 Forces & Equilibrium

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 12t + 10$ m/s. Find when the particle is momentarily at rest.



a. $t = 1$ s only

b. $t = 1$ s or $t = 5$ s

c. $t = 1$ s or $t = 6$ s

d. $t = 2$ s or $t = 5$ s

40 [Challenging] • Target: A well-designed word problem: a train between two stations • Cross-Lesson: 13.01 Newton's Laws

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 3t^2 - 18t + 15$ m/s. Find when the particle is momentarily at rest.

a. $t = 1$ s only

b. $t = 2$ s or $t = 5$ s

c. $t = 1$ s or $t = 5$ s

d. $t = 1$ s or $t = 6$ s

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Constant Acceleration & SUVAT Equations • Cross-Lesson: 14.01 Energy, Work & Power

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 26t + 84$ m/s. Find when the particle is momentarily at rest.

a. $t = 7$ s or $t = 7$ s

b. $t = 6$ s only

c. $t = 6$ s or $t = 7$ s

d. $t = 6$ s or $t = 8$ s

42 [Challenging] • Target: Velocity-Time & Displacement-Time Graphs • Cross-Lesson: 15.01 Representation of Data

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 20t + 48$ m/s. Find when the particle is momentarily at rest.

a. $t = 4$ s only

b. $t = 5$ s or $t = 6$ s

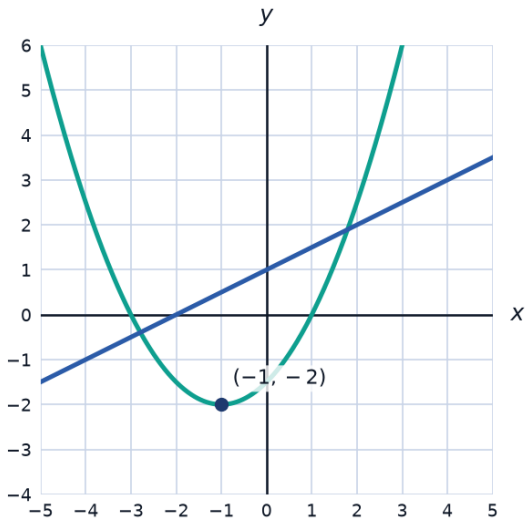
c. $t = 4$ s or $t = 7$ s

d. $t = 4$ s or $t = 6$ s

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Variable Acceleration via Calculus • Cross-Lesson: 16.01 Permutations & Combinations

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 14t + 24$ m/s. Find when the particle is momentarily at rest.



a. $t = 3$ s or $t = 5$ s

b. $t = 3$ s only

c. $t = 4$ s or $t = 4$ s

d. $t = 3$ s or $t = 4$ s

44 [Challenging] • Target: Multi-Stage Motion & Exam Challenges • Cross-Lesson: 17.01 Discrete Random Variables

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 2t^2 - 18t + 28$ m/s. Find when the particle is momentarily at rest.

a. $t = 3$ s or $t = 7$ s

b. $t = 2$ s or $t = 8$ s

c. $t = 2$ s only

d. $t = 2$ s or $t = 7$ s

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Displacement, velocity, and acceleration • Cross-Lesson: 18.01 Binomial Distribution

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = 3t^2 - 21t + 30$ m/s. Find when the particle is momentarily at rest.

a. $t = 2$ s or $t = 5$ s

b. $t = 2$ s only

c. $t = 2$ s or $t = 6$ s

d. $t = 3$ s or $t = 5$ s

© ClipSAT — clipsat.org | All Rights Reserved